Modifying and Correcting Source Data

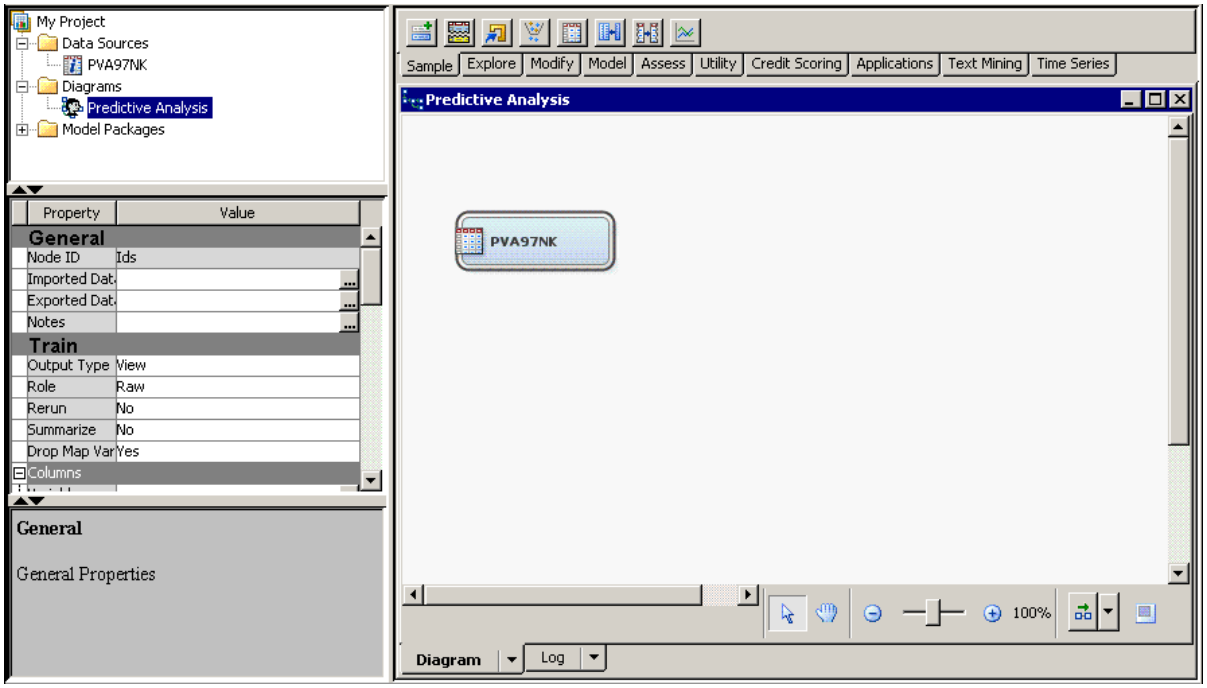
In the **PVA97NK** data set, the **DemMedIncome** variable was seen to have an unusual spike at 0. This phenomenon often occurs in data extracted from a relational database table where 0 or another number is used as a substitute for the value **`missing`** or **`unknown`**. Clearly, having zero income is considerably different from having an unknown income. If you properly use the **income** variable in a predictive model, this discrepancy can be addressed.

This demonstration shows you how to replace a placeholder value for missing with a true missing value indicator. In this way, SAS Enterprise Miner tools can correctly respond to the true, but unknown, value. SAS Enterprise Miner includes several tools that you can use to modify the source data for your analysis. The following demonstrations show how to use the Replacement node to modify incorrect or improper values for a variable:

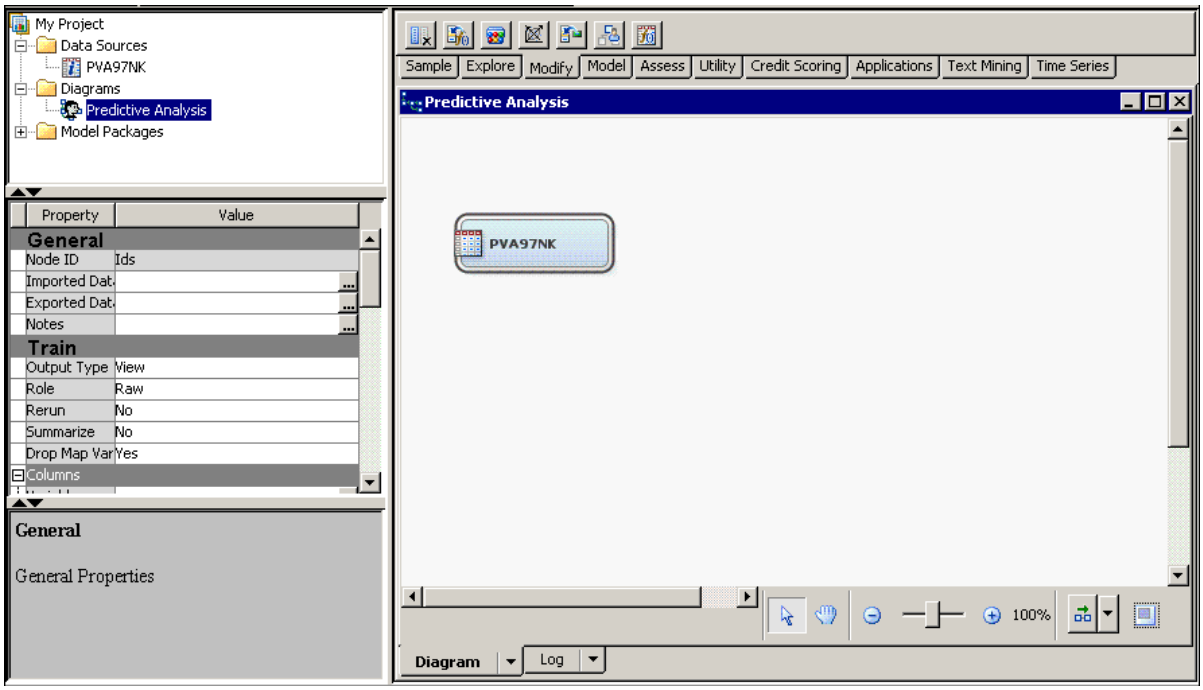
Process Flow Setup

Use the following steps to set up the process flow that will modify the **DemMedIncome** variable.

1. Drag the **PVA97NK** data source to the Predictive Analysis workspace window.

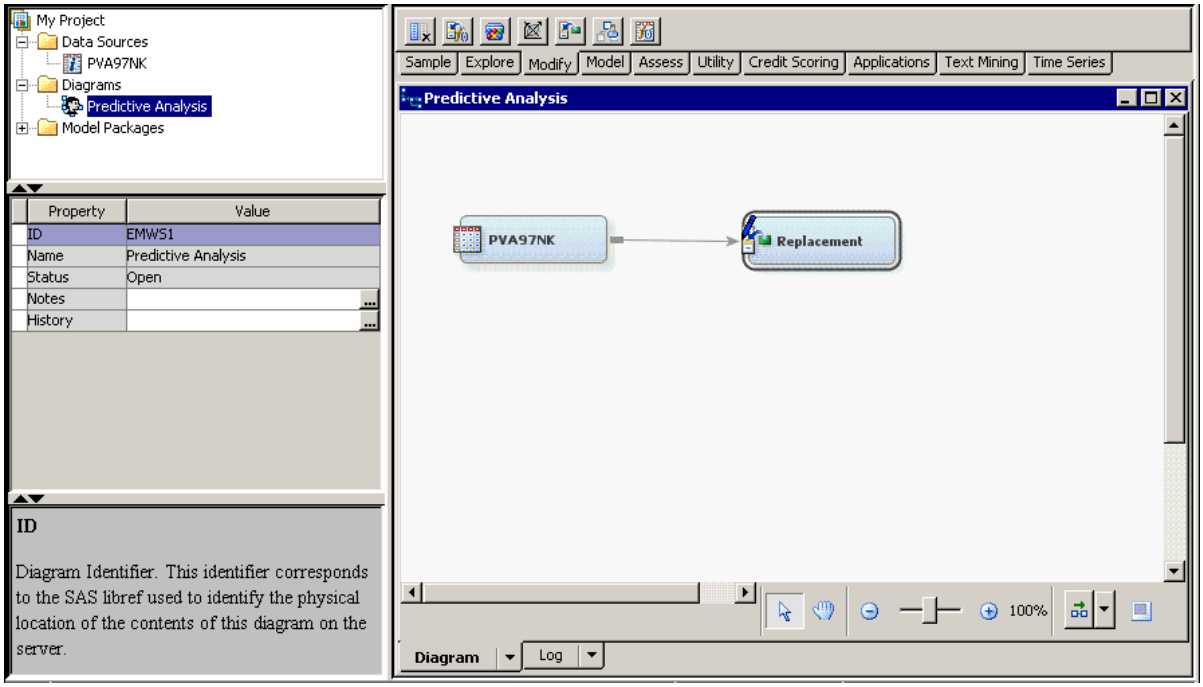


2. Select the **Modify** tab to access the Modify tool group.



3. Drag the **Replacement** tool (third from the right) from the tools palette into the Predictive Analysis workspace window.

4. Connect the **PVA97NK** data to the Replacement node by clicking near the right side of the **PVA97NK** node and dragging an arrow to the left side of the Replacement node.



You created a process flow, which is the method that SAS Enterprise Miner uses to carry out analyses. The process flow, at this point, reads the raw **PVA97NK** data and replaces the unwanted values of the observations. You must, however, specify which variables have unwanted values and what the correct values are. To do this, you must change the settings of the Replacement node.

Changing the Replacement Node Properties

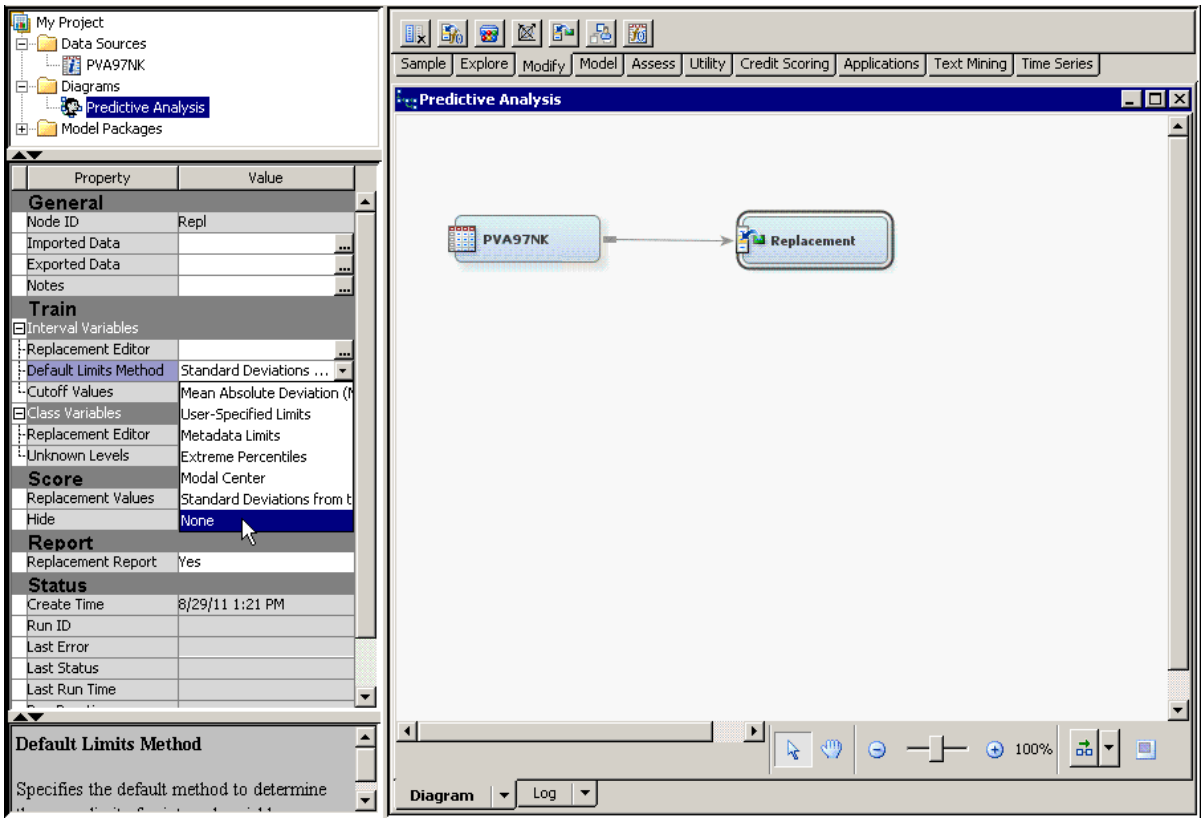
Use the following steps to modify the default settings of the Replacement node.

1. Select the **Replacement** node and examine the Properties panel.

   The Properties panel displays the analysis methods used by the node when it is run. By default, the node replaces all interval variables whose values are more than three standard deviations from the variable mean.

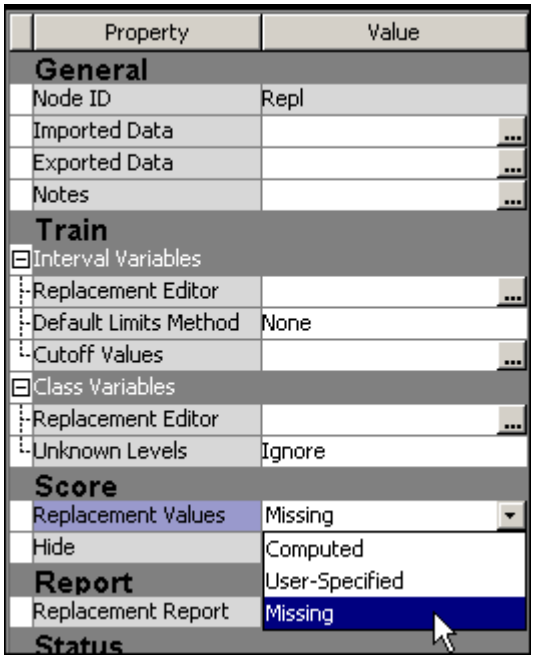
   In this demonstration, you want to replace the value for **DemMedIncome** only when it equals zero. Thus, you need to change the default setting.

2. Select the **Default Limits Method** property and select **None** from the Options menu.



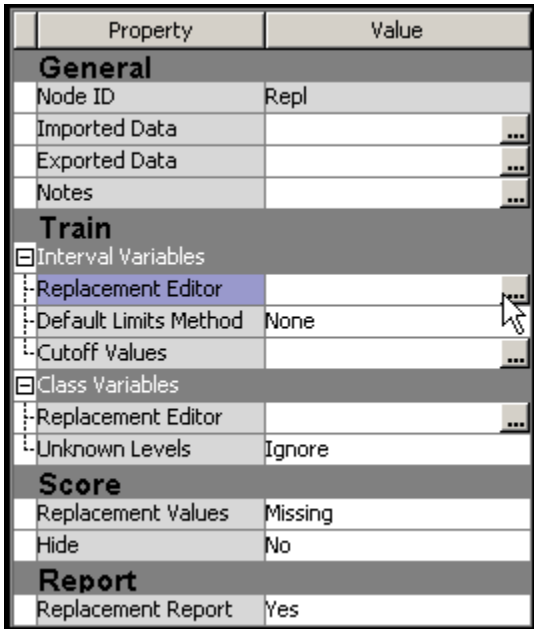
   You want to replace improper values with missing values. To do this, you need to change the Replacement Value property.

3. Select the **Replacement Value** property and select **Missing** from the Options menu.



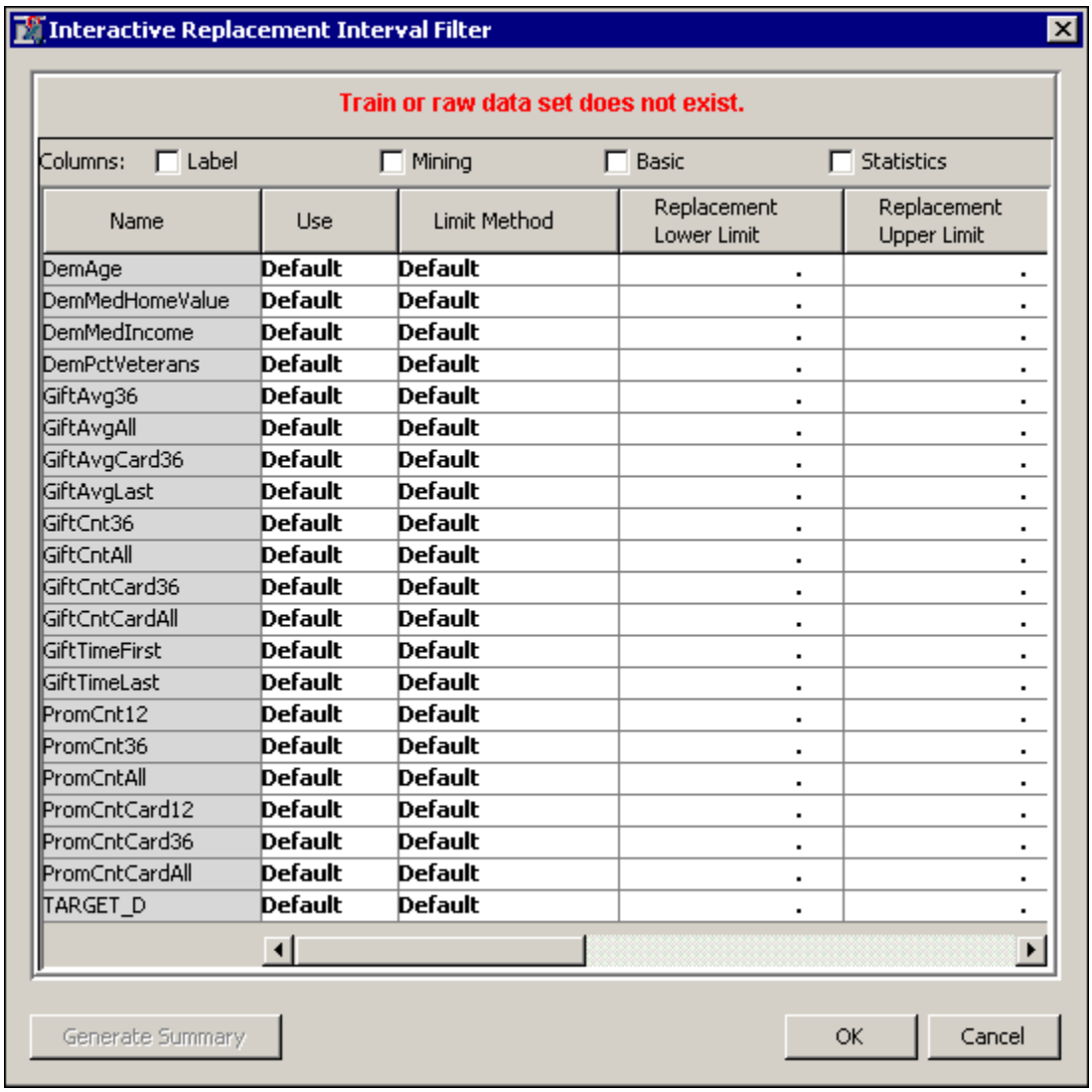
You are now ready to specify the variables that you want to replace.

4. Select ... (Interval Variables: Replacement Editor ellipsis) from the Replacement node properties panel.



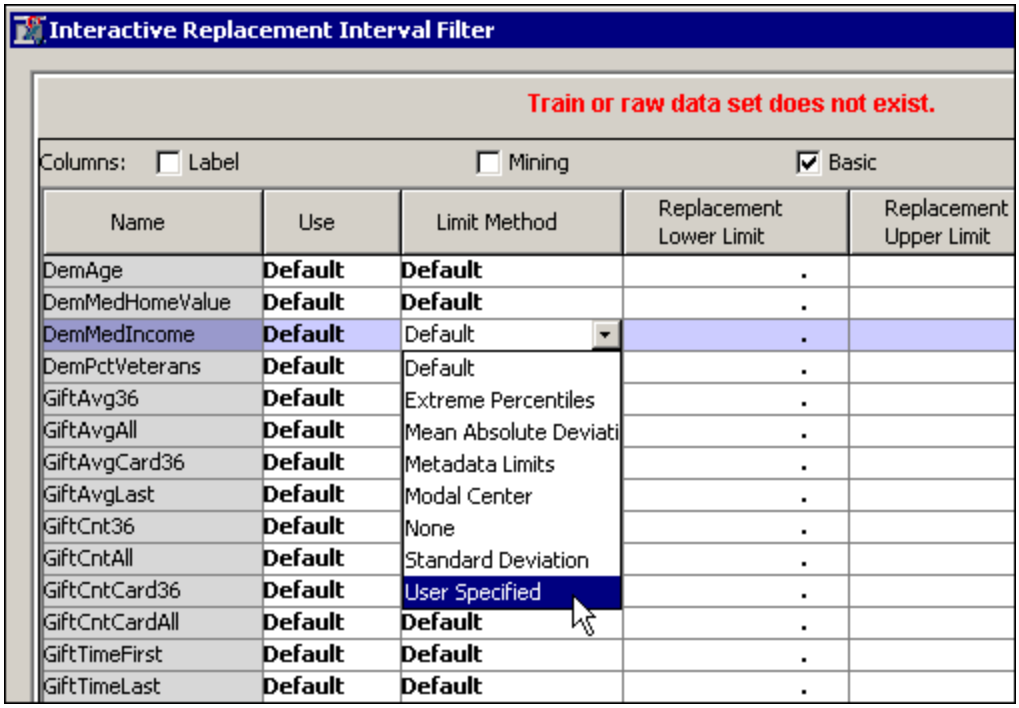
Be careful to open the Replacement Editor for interval variables, ***not*** for class variables.

The Interactive Replacement Interval Filter window appears.

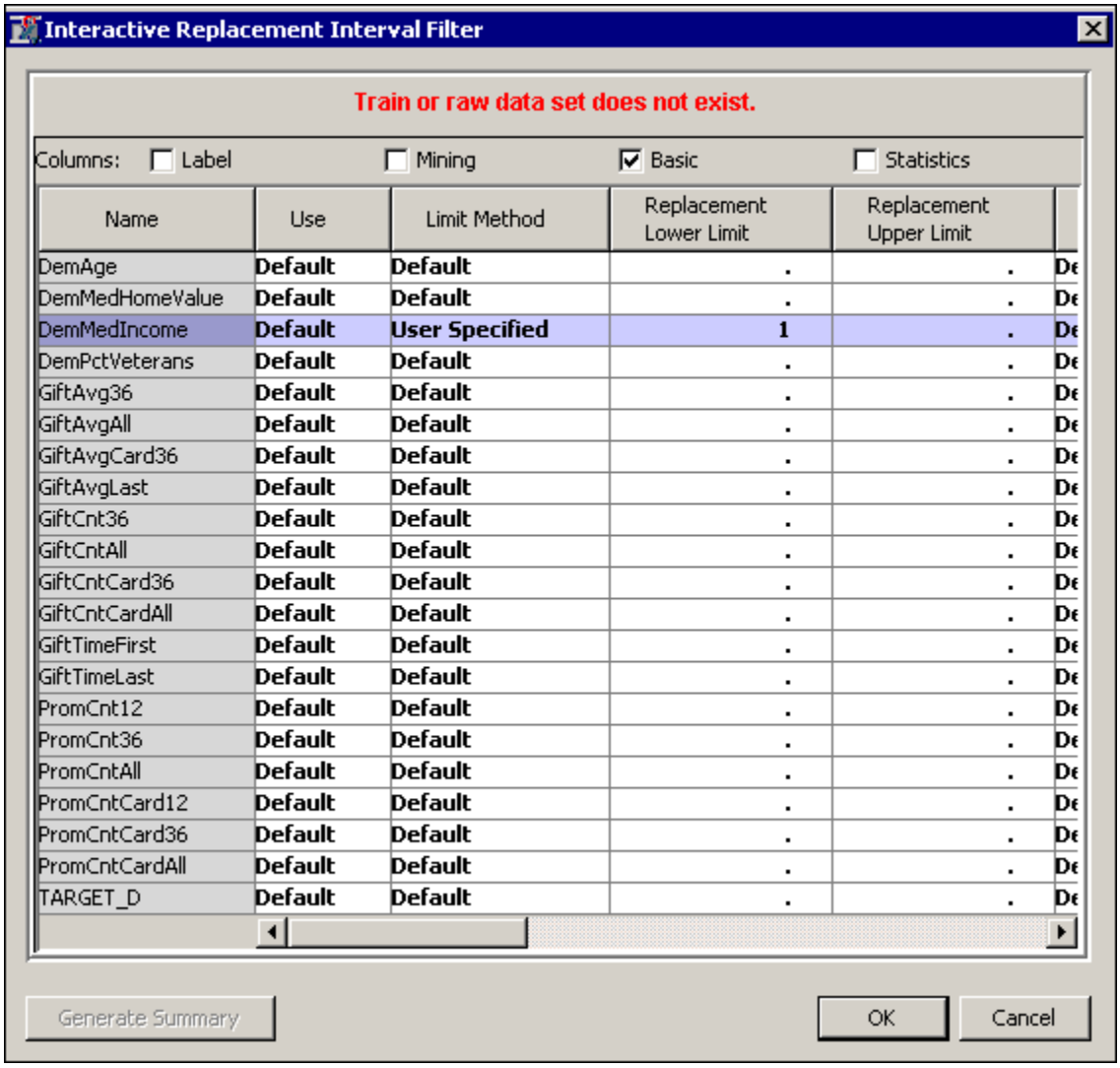


Name	Use	Limit Method	Replacement Lower Limit	Replacement Upper Limit
DemAge	Default	Default	.	.
DemMedHomeValue	Default	Default	.	.
DemMedIncome	Default	Default	.	.
DemPctVeterans	Default	Default	.	.
GiftAvg36	Default	Default	.	.
GiftAvgAll	Default	Default	.	.
GiftAvgCard36	Default	Default	.	.
GiftAvgLast	Default	Default	.	.
GiftCnt36	Default	Default	.	.
GiftCntAll	Default	Default	.	.
GiftCntCard36	Default	Default	.	.
GiftCntCardAll	Default	Default	.	.
GiftTimeFirst	Default	Default	.	.
GiftTimeLast	Default	Default	.	.
PromCnt12	Default	Default	.	.
PromCnt36	Default	Default	.	.
PromCntAll	Default	Default	.	.
PromCntCard12	Default	Default	.	.
PromCntCard36	Default	Default	.	.
PromCntCardAll	Default	Default	.	.
TARGET_D	Default	Default	.	.

5. Select **User Specified** as the Limit Method value for **DemMedIncome**.



6. Type **1** as the Lower Limit value for **DemMedIncome**.



Name	Use	Limit Method	Replacement Lower Limit	Replacement Upper Limit
DemAge	Default	Default	.	.
DemMedHomeValue	Default	Default	.	.
DemMedIncome	Default	User Specified	1	.
DemPctVeterans	Default	Default	.	.
GiftAvg36	Default	Default	.	.
GiftAvgAll	Default	Default	.	.
GiftAvgCard36	Default	Default	.	.
GiftAvgLast	Default	Default	.	.
GiftCnt36	Default	Default	.	.
GiftCntAll	Default	Default	.	.
GiftCntCard36	Default	Default	.	.
GiftCntCardAll	Default	Default	.	.
GiftTimeFirst	Default	Default	.	.
GiftTimeLast	Default	Default	.	.
PromCnt12	Default	Default	.	.
PromCnt36	Default	Default	.	.
PromCntAll	Default	Default	.	.
PromCntCard12	Default	Default	.	.
PromCntCard36	Default	Default	.	.
PromCntCardAll	Default	Default	.	.
TARGET_D	Default	Default	.	.

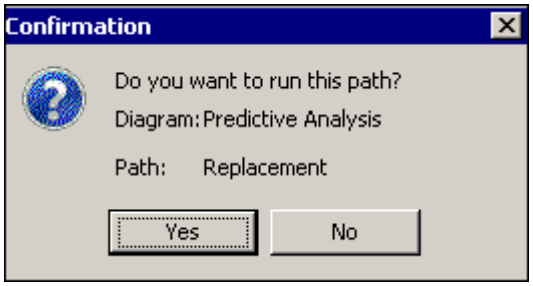
If you use this specification, any **DemMedIncome** values that fall below the lower limit of 1 are set to missing. All other values of this variable do not change.

7. Select **OK** to close the Interactive Replacement Interval Filter window.

Running the Analysis and Viewing the Results

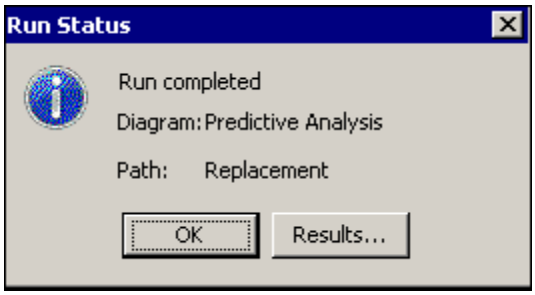
Use these steps to run the process flow that you created.

1. Right-click on the **Replacement** node and select **Run** from the Option menu. A Confirmation window appears and requests that you verify the run action.

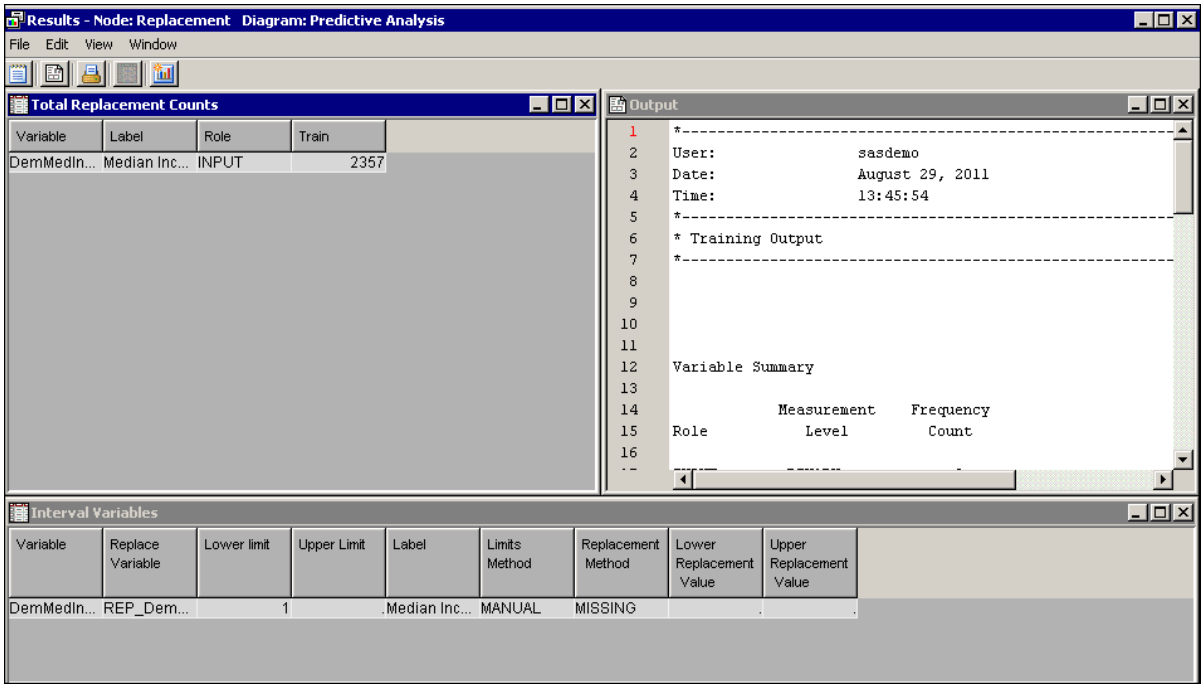


2. Select **Yes** to close the Confirmation window. A small animation in the lower right corner of each node indicates analysis activity in the node.

   The Run Status window opens when the process flow run is complete.



3. Select **Results** to review the analysis outcome. The Results - Node: Replacement Diagram: Predictive Analysis window appears.



The Replacement Counts window shows that 2357 observations were modified by the Replacement node. The Interval Variables window summarizes the replacement that was conducted. The Output window provides more or less the same information as the Total Replacement Counts window and the Interval Variables window (but it is presented as a static text file).

4. Close the Results window.

Examining Exported Data

In a SAS Enterprise Miner process flow diagram, each node takes in data, analyzes it, creates a result, and exports a possibly modified version of the imported data. While the report gives the analysis results in abstract, it is good practice to see the actual effects of an analysis step on the exported data. This enables you to validate your expectations at each step of the analysis.

Use these steps to examine the data exported from the Replacement node.

1. Select the **Replacement** node in your process flow diagram.

2. Select **Exported Data** ⇨ ...

Property	Value
General	
Node ID	Repl
Imported Data	
Exported Data	
Notes	
Train	
Interval Variables	
Replacement Editor	
Default Limits Method	None
Cutoff Values	
Class Variables	
Replacement Editor	
Unknown Levels	Ignore
Score	
Replacement Values	Missing
Hide	No
Report	
Replacement Report	Yes

The Exported Data - Replacement window appears.

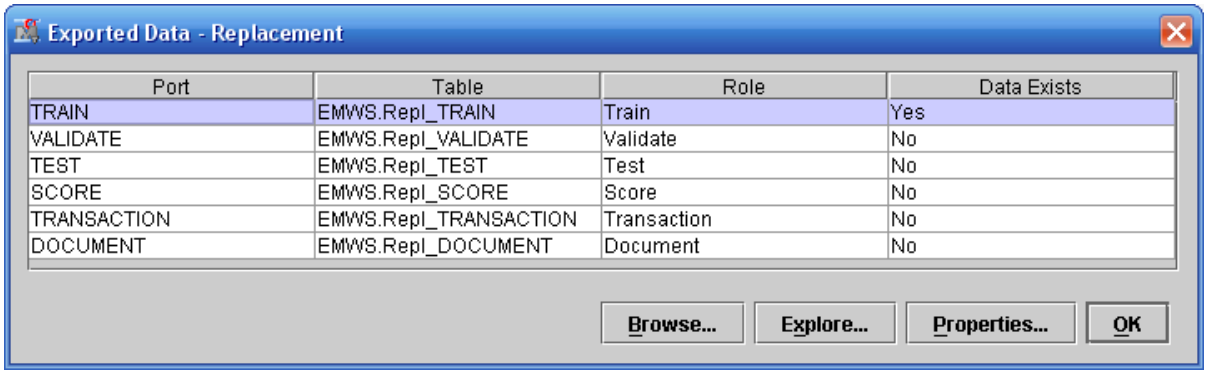
Exported Data - Replacement

Port	Table	Role	Data Exists
TRAIN	EMWS.Repl_TRAIN	Train	Yes
VALIDATE	EMWS.Repl_VALIDATE	Validate	No
TEST	EMWS.Repl_TEST	Test	No
SCORE	EMWS.Repl_SCORE	Score	No
TRANSACTION	EMWS.Repl_TRANSACTION	Transaction	No
DOCUMENT	EMWS.Repl_DOCUMENT	Document	No

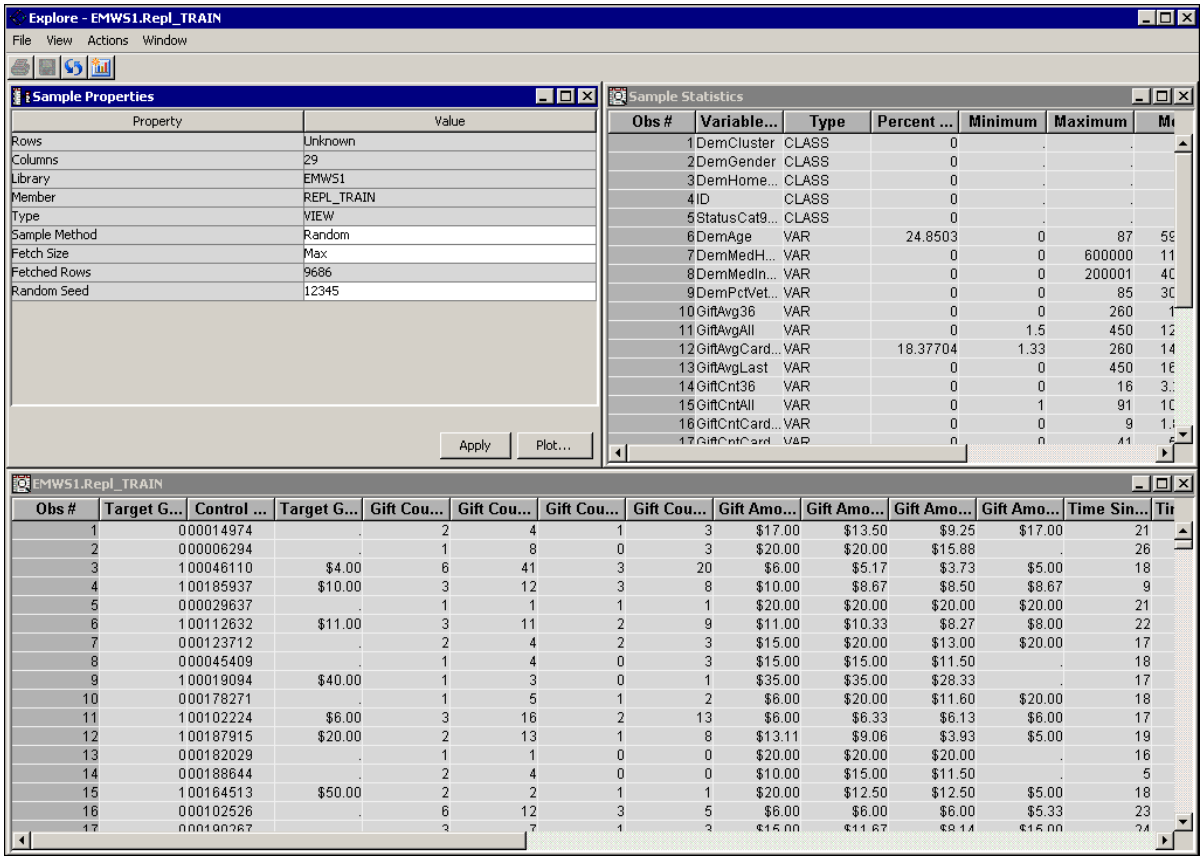
Browse... Explore... Properties... OK

This window lists the types of data sets that can be exported from a SAS Enterprise Miner process flow node. As indicated, only a Train data set exists at this stage of the analysis.

3. Select the **TRAIN** table from the Exported Data - Replacement window.



4. Select **Explore** to access the Explore window again.



5. Scroll completely to the right in the data table.

EMWS1.Repl_TRAIN

Age	Gender	Home O...	Median Home Value Region	Percent Veterans...	Median Income Region	Replacement: Median Income Region
.	F	U	$0	0	$0	.
67	F	U	$186,800	85	$0	.
.	M	U	$87,600	36	$38,750	38750
.	M	U	$139,200	27	$38,942	38942
53	M	U	$168,100	37	$71,509	71509
47	M	H	$253,100	0	$92,514	92514
58	M	H	$234,700	22	$72,868	72868
.	U	U	$207,000	44	$0	.
.	F	U	$137,300	32	$0	.
.	U	U	$180,700	37	$0	.
.	U	U	$143,900	30	$0	.
.	F	U	$134,400	41	$0	.
.	F	U	$68,200	47	$0	.
39	M	H	$115,100	30	$63,000	63000
50	F	H	$0	33	$0	.
45	F	U	$0	37	$0	.
.	M	U	$207,900	48	$0	.

A new column is added to the analysis data: **Replacement: DemMedIncome**. Notice that the values of this variable match the **Median Income Region** variable, except when the original variables value equals zero. The replaced zero value is represented by a period, which indicates a missing value.

6. Close the Explore and Exported Data windows.